

The DevOps Series

Ansible Deployment of LAMP and WordPress

This is the second article in the DevOps series, and covers the installation of a LAMP stack and WordPress, using Ansible.



In this article, we are going to learn how to automate the deployment of a LAMP stack and install WordPress.

LAMP stands for Linux, Apache (a Web server), MySQL (a database) and PHP (server-side scripting). It is a technology stack on which you can deploy different Web applications. We are also going to explore the installation of WordPress, which is free and open source software for creating websites and blogs.

Installing Linux

A Parabola GNU/Linux-libre x86_64 system is used as the host system. An Ubuntu 15.04 image runs as a guest OS using KVM/QEMU. Ansible is installed on the host system using the distribution package manager. You should be able to issue commands from Ansible to the guest OS. For example:

```
$ ansible ubuntu -m ping
ubuntu | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

The `/etc/hosts` file already has an entry for the guest Ubuntu VM.

```
192.168.122.250 ubuntu
```

On the host system, we will create a project for our playbooks. It has the following directory structure:

```
ansible/inventory/kvm/
  /playbooks/configuration/
  /playbooks/admin/
```

An 'inventory' file is created inside the `inventory/kvm`

folder that contains the following:

```
ubuntu ansible_host=192.168.122.250 ansible_connection=ssh
ansible_user=xetex
```

Installing Apache

We will first install and test the Apache Web server on the guest Ubuntu system. Let's then create a `playbooks/configuration/apache.yml` file with the following content:

```
---
- name: Install Apache web server
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [web]
  tasks:
    - name: Update the software package repository
      apt:
        update_cache: yes
    - name: Install Apache
      package:
        name: "[{ item }]"
        state: latest
        with_items:
          - apache2
    - wait_for:
        port: 80
```

On the Ubuntu guest system, the playbook runs `apt-get` and then installs the `apache2` package. The playbook finishes after the server has started, and is listening on port 80. Open a terminal, enter the `ansible/` folder, and execute the